RIDDHESH SAWANT

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TECHNICAL SKILLS

Core Competencies: Statistical Modeling, Machine Learning, Time Series Forecasting, Causal Inference, ETL, Natural Language Processing, Database Management, Distributed Computing, Data Visualization, Data Structures & Algorithms, Convex Optimization, LLM Serving Systems

Programming Languages: Python, R, PySpark, SQL, C++, Html, MATLAB

Frameworks : PyTorch, TensorFlow, Keras, NumPy, Pandas, Scikit-learn, Matplotlib, Gurobi, OpenAI APIs, REST APIs

Tools & Platforms: AWS, Databricks, Docker, Git, Airflow, Snowflake, Tableau, Kubernetes, MLflow, FastAPI, Kafka, CI/CD, Retool, Mode, MS Excel

EXPERIENCE

Faire — **San Francisco, CA**

Data Science Intern — June 2025 - Present

- Built and deployed a retailer sales estimation agent using **deep research, vector embeddings**, and transaction data (**±15% error for 80% of retailers**). Added a calibration layer for bias mitigation and monthly auto-retraining via **Airflow + AWS Batch**, reducing bad debt risk by **12%**.
- Developed an AI fraud detection and retailer verification system with **GPT-powered multi-agent workflows**, processing 1,000+ applications per batch. Achieved **95%+ accuracy**, prevented **$500K+ fraud losses**, and cut review time from **30 minutes to 5 seconds** on scalable cloud infrastructure

CRED — **Bengaluru, India**

Data Scientist II — September 2022 - August 2024

- Engineered a dynamic pricing framework to optimize interest rates, boosting **portfolio conversion by 17%** and driving **$12M in incremental monthly disbursals**; with **CatBoost (AUC 0.83)** and **constraint optimization**.
- Developed a contrastive learning framework using a **Transformer-based sequence encoder (SASRec + SimCLR-style loss)** on user traversal data, learning embeddings that clustered high-LTV power users vs low-engagement users. Achieved **92% precision** in identifying high-value cohorts, enabling targeted credit scaling and boosting profitability by **7%**.
- Developed a **Bayesian time-series forecasting system (Orbit)** to predict credit disbursal demand and default risk for a **$200M portfolio**, achieving **94% forecast accuracy** and reducing default risk by **6%**; automated retraining and reporting, saving **20+ hours monthly**.
- Collaborated with product managers and risk teams to align ML outputs with credit policy, improving adoption across business.
- Deployed ML models as **real-time microservices** using **AWS + Docker + Kubernetes**, reducing scoring latency from **250ms to 50ms** and scaling to **500K+ daily requests**.

Data Scientist — June 2021 - September 2022

- Built a personalized homepage recommendation engine with gradient boosting and user segmentation, increasing **loan application clicks by 30%**.
- **Increased approval rates by 5%** with a liability validation model (**random forest, AUC 0.84 + linear optimization**).
- Built a **real-time monitoring platform** with drift detection, residual analysis, and Slack/email alerts, integrated with the **feature store** to ensure consistency across training and serving. Onboarded **15+ production models**, cutting incident response time by **80%**

Data Scientist Intern — January 2021 - June 2021

- **Improved offer conversion rate by 15%** by designing a loan uptake **propensity model** using xgboost(0.78 AUC)
- Engineered **in-app/external communication triggers**, contributing to **10%+ of monthly loan disbursals**.

EDUCATION

University of Washington — Seattle, WA

Master of Science (M.S.), Data Science — September 2024 - March 2026

Birla Institute of Technology and Science Pilani — Goa, India

Bachelor of Engineering (B.E.), Electronics and Communication — August 2017 - May 2021

PUBLICATIONS AND PROJECTS

Novel EEG Features for Consumer Emotion Prediction using Correlation-Based Subset Selection

- Presented at AMC IMX 2022 Conference

MICCAI UNICORN Medical AI Challenge - UW KurtLab — **March 2025 - Present**

- Fine-tuned open-source healthcare models (AlphaMed, UltraMed, MedIron) and general-purpose transformers (LLaMA, RoBERTa) via Hugging Face APIs, applying cross-modal learning with efficient transformer backbones under compute limits for pathology and radiology tasks.

Underwater Acoustic Monitoring for Marine Conservation (Orcasound Project) — **February 2025 - June 2025**

- **Developed scalable pipelines** to process 3,000+ daily hydrophone recordings and deployed real-time orca monitoring dashboards, accelerating noise trend analysis by **5x** and expanding data access by **300%** for 10+ research projects.